

## Highlights

### **Explainable production-adjusted electrical-energy baselines for industrial feed pelleting**

Vladimir Sousa Santos<sup></sup>, Juan José Cabello Eras<sup></sup>, Eder Elías Herrera Rodiño, Francisco Manuel Arrabal-Campos<sup></sup>

- Production-adjusted electricity baseline from 2,745 real feed-pelleting records
- Frozen Ridge model cuts test MAE by 14.42% relative to line-specific SEC
- Completed same-line histories, TFT, N-HiTS, TimesFM 2.5 and Chronos-2 add no accuracy
- Exact additive decomposition traces each kWh estimate to recorded production inputs
- Fixed 90%/95% residual intervals cover 85.96% and 93.23% of later test records

# Explainable production-adjusted electrical-energy baselines for industrial feed pelleting

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## Abstract

Electricity use in industrial feed pelleting varies with the quantity produced, the product, the pelleting line and the batch organization, so neither raw consumption nor a single specific energy consumption (SEC) figure says much about energy performance. This retrospective study developed and chronologically evaluated an explainable, production-adjusted electricity baseline and asked whether histories of completed operations improve it. The data are 2,745 records from three pelleting lines, 2,742 with positive recorded energy; the inputs are dosed mass, dosing-batch count, product characteristics, pelleting-line identity and calendar variables. Seven methods were compared on 800 chronologically ordered development targets; the selected Ridge regression was then frozen and evaluated against line-specific SEC on 798 later records. Eight current-input, history-enabled, temporal and pretrained configurations were also compared on 584 matched targets with 16 completed same-line operations as context. Ridge reduced test MAE from 61.29 to 52.45 kWh, 14.42% below line-specific SEC; this is lower prediction error, not an energy

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saving. On the matched population the Temporal Fusion Transformer reached 53.65 kWh against 53.55 kWh for current-input Ridge, with no evidence of paired improvement, and the tested N-HiTS, TimesFM 2.5 and Chronos-2 configurations were less accurate. Performance was most consistent on Pellet 2 and Pellet 3; Pellet 1 also improved over SEC but remained systematically underestimated. Fixed 90% and 95% residual intervals covered 85.96% and 93.23% of the test records. Ridge provides a transparent retrospective baseline; prospective use requires verified inputs and measurement boundaries, line-specific acceptance criteria and validation on a new consecutive period.

**Keywords:** feed pelleting, industrial energy baseline, production adjustment, chronological validation, prediction intervals, explainable Ridge regression

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## Nomenclature

### Symbols

|                                       |                                                                                     |
|---------------------------------------|-------------------------------------------------------------------------------------|
| $y_i, E_i$                            | pelleting electrical energy of record $i$ (kWh)                                     |
| $\hat{y}_i$                           | point estimate of $y_i$ (kWh)                                                       |
| $m_i$                                 | dosed mass of record $i$ (t)                                                        |
| $b_i$                                 | number of dosing batches of record $i$                                              |
| $\ell_i, c_i, p_i, s_i$               | pelleting line, product subfamily, presentation and bag/bulk category of record $i$ |
| $h_i, d_i$                            | origin hour (h) and weekday (0–6) of record $i$                                     |
| $t_i$                                 | analytical origin of record $i$ : earliest recorded stage start                     |
| $a_j$                                 | recorded pelleting completion of record $j$                                         |
| $\Delta_i$                            | recorded pelleting duration of record $i$ (h)                                       |
| $P_i^{\text{avg}}$                    | duration-based average-power proxy of record $i$ (kW)                               |
| $z_i$                                 | standardized and one-hot encoded input vector of record $i$                         |
| $\hat{b}, \hat{\beta}$                | fitted intercept and coefficient vector of the Ridge model                          |
| $\hat{s}_\ell$                        | training-period specific energy consumption of line $\ell$ (kWh/t)                  |
| $\mathcal{T}, \mathcal{P}$            | training set and partition interval                                                 |
| $\mathcal{H}_i^{(16)}, \mathcal{H}_i$ | 16-record operating history and recent-residual history of target $i$               |
| $C_{ig}$                              | contribution of variable group $g$ to the estimate of record $i$ (kWh)              |
| $\delta_i$                            | shrunk recent-residual correction (kWh)                                             |
| $q_\alpha$                            | calibrated half-width at miscoverage level $\alpha$ (kWh)                           |
| $L_i, U_i$                            | lower and upper interval limits of record $i$ (kWh)                                 |
| $r_i^{\text{excess}}, B_i$            | signed excess over the estimate (kWh) and out-of-interval flag                      |

### *Abbreviations*

|                 |                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------|
| EnB, EnPI       | energy baseline, energy performance indicator                                                              |
| IPMVP           | International Performance Measurement and Verification Protocol                                            |
| MAE, RMSE, WAPE | mean absolute error, root mean squared error, weighted absolute percentage error                           |
| IS              | interval score                                                                                             |
| MLP             | multilayer perceptron                                                                                      |
| SEC             | specific energy consumption (kWh/t)                                                                        |
| TFT             | Temporal Fusion Transformer                                                                                |
| N-HITS          | Neural Hierarchical Interpolation for Time Series                                                          |
| XReg            | covariate-regression adapter of TimesFM                                                                    |
| H1, H2, H3      | post-hoc extensions: line-dependent mass effect, product-dependent mass effect, recent-residual correction |
| GPU, CPU        | graphics and central processing unit                                                                       |

## **1. Introduction**

Improving energy performance in manufacturing means comparing the electricity an operation actually used with what it should have used under comparable conditions [1]. Food manufacturing is a case in point. Its electricity goes to drives, refrigeration, pumping, mixing, cooling, conveying and process control; food processing in the United States has been estimated to consume about 75 TWh of electricity, and electricity accounts for roughly 30% of the reported energy use of UK food and beverage manufacturing [2, 3]. Figures of that size only become actionable when consumption is read against production activity and operating conditions rather than in absolute terms. Energy management standards make the point explicit: ISO 50001 asks organizations to judge energy performance against a baseline normalized for the relevant variables, and ISO 50006 and the IPMVP describe how such baselines and performance indicators are built and used [4, 5, 6].

Compound-feed manufacturing is a representative setting because its production stages depend on electromechanical equipment whose demand varies with volume and process conditions. Sánchez et al. [7] assessed motor efficiency and established energy baselines and performance indicators for a balanced-feed plant, reporting about 10 kWh/t at 500 t/day and an estimated saving potential of 364,000 kWh/year. Their work shows the value of normalizing consumption by production level, but the baseline was formulated at an aggregated process level; it did not deal with heterogeneous operation-level records in which product, pelleting line, batch organization and other descriptors change at the same time.

From a process perspective, pelleting is intrinsically multivariable. Pellet quality, conditioning, feed composition, steam application, temperature, particle characteristics and operating conditions all influence pelleting behaviour and mechanical energy demand [8, 9, 10, 11, 12]. These studies establish the physical basis for energy differences between operations, but their aims lie in material properties, process behaviour or pellet quality rather than in building and chronologically testing an operation-level electricity baseline from real plant records.

Specific energy consumption (SEC), usually in kWh/t, is simple and interpretable, but a single value cannot separate the effects of product mix, pelleting-line assignment, processed mass, batch structure and operating regime. Regression-based and data-driven baselines have therefore been used to estimate expected consumption under comparable conditions and to support measurement and verification [13, 14, 15, 16, 7]. They demonstrate why expected energy use should be conditioned on production variables, but they have not been tested on heterogeneous feed-pelleting operations under a strict chronological protocol.

Data-driven modelling offers several ways to extend such baselines. Regularized linear regression gives stable, interpretable estimates when predictors are correlated [17], and tree ensembles and neural networks can capture nonlinearities and interactions [18]. Closest to our setting, Ekwaro-Osire et al. [19] compared linear regression, tree-based methods and a multilayer perceptron on about 1,100 production batches with 67 input characteristics from an industrial compound-feed pelleting process and found that more complex models did not necessarily reduce error, which supports transparent statistical models as strong references. That study, however, relied on shuffled cross-validation and did not consider chronologically available histories of completed operations, pretrained temporal models, predictive uncertainty or record-level traceability within one comparison.

Temporal and pretrained time-series models add a further perspective. The Temporal Fusion Transformer (TFT) combines recurrent layers, variable selection, gating and attention [20]; N-HiTS represents temporal structure through hierarchical interpolation [21]; and foundation models such as TimesFM and Chronos transfer representations learned from large time-series collections to new tasks [22, 23]. Industrial pelleting records differ from regularly sampled series, though: consecutive completed operations may have different masses, products, lines, start times and durations. The relevant question is whether a history restricted to operations completed before a target carries information beyond the descriptors of the target itself.

That question is inseparable from validation design. Random or shuffled partitions mix operating regimes across time and inflate apparent performance [24],

so chronological evaluation is required whenever assessment should reflect the temporal direction of industrial use [25, 26, 27]. The requirement tightens when historical outcomes become inputs, because only operations whose results were available before the target origin may enter its context. Development, calibration and testing must therefore stay temporally separated, and the value of history must be judged against a current-input model on identical eligible records.

Interpretability and uncertainty matter as well. Explainable-AI research has shown the value of local and global explanations alongside limitations in robustness, standardization and generalization [28, 29]. An industrial baseline should let every estimate be traced to the recorded production variables without presenting model terms as causal effects. Prediction intervals are needed because low average error does not guarantee the same reliability on every line or in every period; calibration and width assessment provide the framework [30, 31, 32, 33, 34], and subgroup evaluation is essential when decisions are taken line by line.

Previous work has thus addressed the physical drivers of pelleting energy demand, energy performance indicators, production-adjusted baselines, machine-learning prediction, temporal forecasting, explainability and uncertainty quantification, but largely in isolation. It remains unclear whether an explainable production-adjusted baseline beats line-specific SEC for heterogeneous feed-pelleting operations under strict chronological validation, whether completed same-line histories add predictive value beyond current-operation variables, and whether regression, nonlinear, temporal and pretrained models can be compared on the same eligible records while interpretability and uncertainty are assessed together.

This study develops and chronologically evaluates explainable, production-adjusted electricity baselines from real operational records of three industrial feed-pelleting lines. Simple operational references, regularized regression, nonlinear ensembles, locally trained temporal models and pretrained time-series configurations are evaluated in one framework. The validated scope is retrospective assessment at the closure of recorded operations, when consolidated production descriptors and outcomes are available; prospective monitoring is a later application that requires confirming predictor availability at the intended origin and validating the full procedure on a new consecutive period. The main contributions are:

1. production-adjusted baselines for heterogeneous operation-level feed-pelleting records, compared with line-specific SEC references;
2. an assessment of whether completed same-line operating histories add predictive information beyond current-operation variables;

3. chronological development and testing that expose model stability across later periods and individual pelleting lines;
4. global and line-specific residual intervals that quantify predictive uncertainty; and
5. record-level interpretability through an exact additive decomposition of the Ridge predictor.

For a plant, the framework separates the ordinary variability of electricity use that follows from production conditions from deviations that deserve a closer look. It supports energy-performance monitoring, identification of atypical operations, comparison among pelleting lines and prioritization of operational reviews. Because the baseline is production-adjusted, chronologically evaluated and accompanied by uncertainty and record-level interpretability, it can also serve as a statistical component of energy-management platforms or industrial digital twins once prospective input availability and measurement boundaries have been verified. Section 2 describes the records, models and evaluation protocol, Section 3 reports the results, Section 4 discusses their implications and limits, and Section 5 concludes.

## 2. Materials and methods

### 2.1. Study design

This is a retrospective observational model-development study. It uses recorded operations from three industrial feed-pelleting lines to build and evaluate production-adjusted baselines for electrical energy, and it estimates the expected energy of completed operations rather than forecasting a continuously sampled power signal. The observational unit is an operation represented by one row of the plant database.

The primary evaluation develops and selects models on temporally ordered validation periods, evaluates the selected model, frozen, against operational reference baselines in an independent chronological test period, and assesses predictive performance and residual-interval coverage overall and by pelleting line. Complementary analyses examine the predictive value of short histories of completed same-line operations, locally trained temporal architectures and pretrained time-series models, line- and product-specific mass interactions, a recent-residual correction, rolling updates of the interval widths, and numerical agreement between computational implementations. These analyses characterize alternative modelling hypotheses; none of them replaces the model frozen for the primary test. Figure 1 summarizes the workflow.

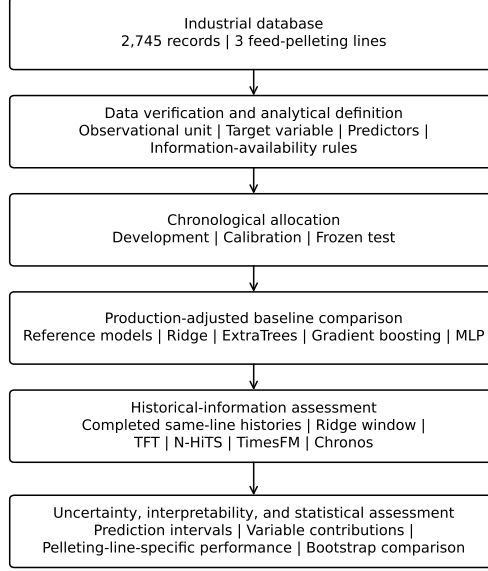


Figure 1: Methodological workflow. The primary evaluation comprises chronological model development, frozen test evaluation and line-specific assessment; the complementary analyses examine short-context temporal models, line- and product-specific interactions, recent-residual correction, rolling interval updates and computational diagnostics.

## 2.2. Industrial data source and observational unit

The source workbook holds 2,745 records and 32 named variables from three feed-pelleting lines: 730 from Pellet 1, 1,148 from Pellet 2 and 867 from Pellet 3. Recorded stage starts run from 2 January to 1 April 2025, and the last pelleting operation was completed on 2 April 2025. The plant information system generated each row as a consolidated record of one production operation, with dosed mass, dosing-batch count, product classifications, stage timestamps, pelleting electrical energy, steam-related variables and operational summaries; the field KWH PELET is the energy value the system assigned to that row.

The database contains no verified production-order identifier linking every recorded stage to a single physical batch, and no separate history of meter readings from which the row-level energy could be reconstructed. A “recorded production operation” is therefore one row of the workbook, not an independently verified physical batch, and source-row identifiers are preserved throughout for traceability. Records with overlapping timestamps were kept when they had distinct identifiers, because overlap alone is not evidence of duplication; overlaps were characterized

by line and treated as a data-quality descriptor when interpreting the chronological analyses.

### 2.3. Target variable, predictors and preprocessing

The response  $y_i$  is KWH PELET, the electrical energy attributed to pelleting record  $i$  in kWh. It excludes the separately recorded electricity of milling and mixing and the thermal energy supplied as steam. Dosed mass  $m_i$  comes from PESO DOS (kg) converted to tonnes. Energy, a duration-based average-power proxy and specific energy consumption are defined as

$$E_i = y_i \text{ [kWh]}, \quad P_i^{\text{avg}} = \frac{y_i}{\Delta_i} \text{ [kW]}, \quad \text{SEC}_i = \frac{y_i}{m_i} \text{ [kWh/t]}, \quad (1)$$

where  $\Delta_i$  is the recorded pelleting duration in hours.  $P_i^{\text{avg}}$  is derived from recorded energy and duration and is not a power measurement.  $E_i$  is the modelling target; SEC is used only to construct the operational reference baselines.

The analytical origin  $t_i$  is the earliest recorded start among milling, mixing and pelleting. It assigns records to chronological partitions and decides whether a historical outcome was completed before a target. The current-record predictors are dosed mass, dosing-batch count, pelleting-line identity, product subfamily, presentation, bag/bulk category, origin hour and origin weekday (Table 1). With  $b_i$  the number of dosing batches,  $h_i$  the origin hour and  $d_i \in \{0, \dots, 6\}$  the weekday indexed from Monday, the numerical input vector is

$$x_i^{\text{num}} = \left( m_i, b_i, \sin \frac{2\pi h_i}{24}, \cos \frac{2\pi h_i}{24}, \sin \frac{2\pi d_i}{7}, \cos \frac{2\pi d_i}{7} \right). \quad (2)$$

The six numerical coordinates are standardized with parameters estimated from the corresponding training data only, and the four categorical variables are one-hot encoded with training vocabularies, giving 22 columns in the Ridge design matrix. A category absent from a training period produces no active indicator in its field. Because grouped Ridge contributions depend partly on this coding, they are read as components of the fitted representation, not as invariant physical effects.

The primary analysis is retrospective and conditional on the consolidated production characteristics recorded for each operation. Current-record durations, stage-specific energy values and the complete steam, temperature and electrical-current summaries are excluded from the predictors. For historical outcomes, recorded pelleting completion is taken as the availability time. A prospective implementation would have to restrict the model to descriptors verified as available at the intended prediction origin.

Table 1: Model inputs: source fields, preprocessing, recorded support and conditions for prospective use. Original Spanish field names are retained for traceability; counts and ranges refer to all 2,745 source records.

| Input / units                                | Source field and preprocessing                    | Interpretation and recorded support                                                                                                             | Prospective use                                                       |
|----------------------------------------------|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| Dosed mass<br>$m_i$ / t                      | PESO DOS (kg);<br>divided by 1,000                | Material entering the operation;<br>2.69–36.78 t, median 14.98 t;<br>the mass denominator used<br>throughout                                    | Possible if the mass<br>is available at the<br>intended origin        |
| Dosing<br>batches $b_i$ /<br>count           | N_BACHES_DOS;<br>numerical                        | One to ten, median five;<br>describes how the mass is<br>aggregated and is correlated<br>with it; not a physical-cycle<br>identifier            | Possible when the<br>batch plan or<br>completed count is<br>available |
| Pelleting line<br>$\ell_i$ / category        | PELLT; one-hot                                    | Pellet 1, 2 and 3: 730, 1,148<br>and 867 records; the separate<br>field LINEA holds product<br>families                                         | Possible when the<br>line assignment is<br>confirmed                  |
| Product<br>subfamily $c_i$ /<br>category     | SUBLINEA; one-hot                                 | Nine labels spanning poultry,<br>swine, cattle, equine and rabbit<br>products; a classification, not a<br>measured formulation                  | Possible when the<br>product<br>classification is<br>known            |
| Presentation<br>$p_i$ / category             | PRESENTACION;<br>one-hot                          | PELETIZADO (pelleted): 2,065;<br>CROMBELIZADO (crumbled):<br>680; may also reflect product<br>allocation                                        | Possible when the<br>presentation is<br>known                         |
| Bag/bulk $s_i$ /<br>category                 | DES_PRODUCTO_<br>TIPO; one-hot                    | ENSAQUE (bagged): 2,420;<br>GRANEL (bulk): 325; the<br>analytical alias<br>salida_programada<br>identifies output form, not a<br>departure time | Possible when the<br>output form is<br>known                          |
| Origin hour /<br>dimensionless<br>pair       | Earliest recorded stage<br>start; sine and cosine | Two coordinates avoid a<br>discontinuity between 23:59<br>and 00:00; seconds are not used                                                       | Available at or<br>immediately after<br>the origin                    |
| Origin<br>weekday /<br>dimensionless<br>pair | Same origin; sine and<br>cosine                   | Two coordinates represent<br>recurring weekly associations,<br>not operational state                                                            | Available at the<br>origin                                            |

#### 2.4. Data integrity and temporal consistency

All 32 source fields were reconciled with the analytical table across the 2,745 records; the 87,840 cell-level comparisons satisfied the predefined numerical and whitespace tolerances, confirming that the analytical copy matches the workbook. Three records (0.11%) with KWH PELET=0 were excluded from supervised modelling because they provide no positive target; their cause could not be established independently and no targets were imputed. Table 2 lists the main integrity and consistency checks; the counts describe different characteristics of possibly the same records and are not mutually exclusive.

Table 2: Data-integrity and temporal-consistency checks.

| Check                              | Count | Analytical treatment                                                     |
|------------------------------------|-------|--------------------------------------------------------------------------|
| Source records                     | 2,745 | Reconciled with the analytical table.                                    |
| Records with zero pelleting energy | 3     | Excluded from supervised targets; not imputed.                           |
| Imputed auxiliary values           | 15    | Retained in the source copy; affected variables were not model inputs.   |
| Within-line interval overlaps      | 351   | Retained as distinct source records and characterized by pelleting line. |
| Pellet 1 overlaps                  | 75    | Included in the line-level overlap count.                                |
| Pellet 2 overlaps                  | 131   | Included in the line-level overlap count.                                |
| Pellet 3 overlaps                  | 145   | Included in the line-level overlap count.                                |
| Test records exceeding 24 hours    | 5     | Retained in the primary test population.                                 |
| Pre-test eligibility exclusions    | 25    | Fourteen development and eleven calibration candidates.                  |

Median recorded pelleting durations are about 145, 96 and 123 minutes on Pellet 1, Pellet 2 and Pellet 3. Milling completion falls after pelleting start in 528 records and mixing completion in 376, and the two sets overlap. These patterns are treated as characteristics of the plant’s consolidated records rather than as exclusion criteria. The dataset contains event-level summaries, not dense power trajectories or verified active/inactive state labels; a separately supplied set of synthetic ten-second series was therefore kept apart from the industrial evidence.

#### 2.5. Chronological evaluation design

Records are assigned by analytical origin to development (origins before 22 February 2025), calibration (22 February to 5 March 2025) and testing (from 6

March 2025); the last operation completes on 2 April 2025 (Figure 2). Development and calibration eligibility require a positive recorded outcome, a valid duration, complete required inputs and completion before the next partition boundary, which retains 1,542 of 1,556 development candidates and 380 of 391 calibration candidates; the 14 and 11 exclusions are the 25 pre-test exclusions of Table 2. All 798 test candidates are retained, including the five records longer than 24 hours.

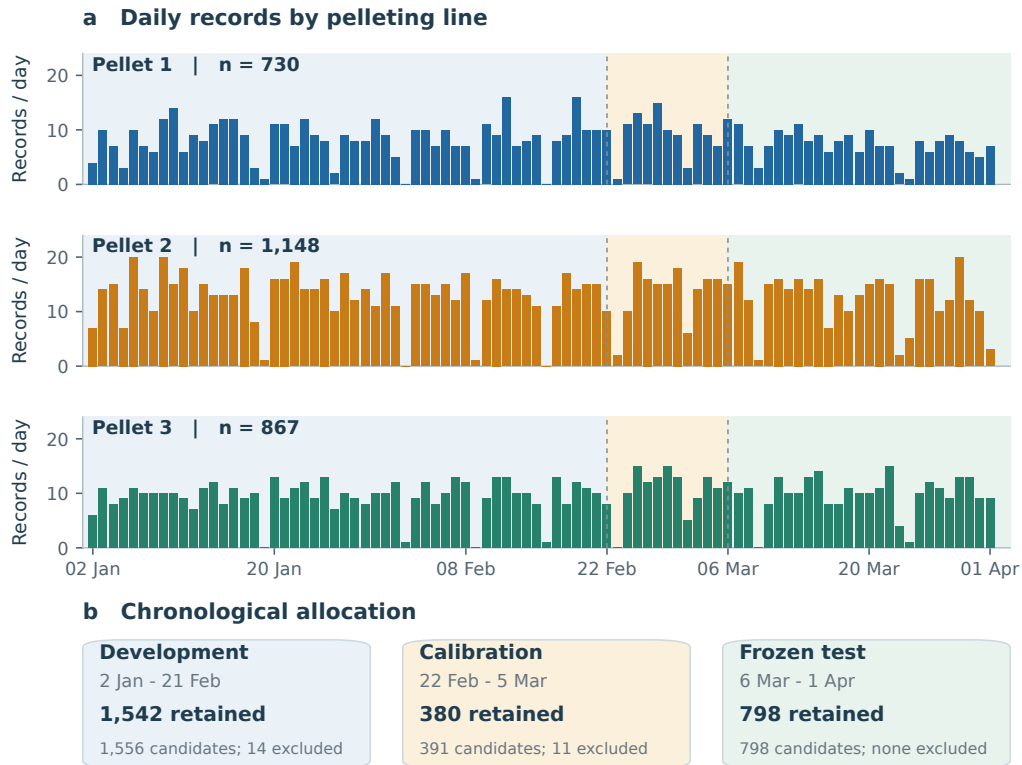


Figure 2: Chronological allocation by pelleting line and partition. (a) Daily record counts for the three lines; background shading marks the development, calibration and frozen-test partitions, and dashed lines mark the development–calibration boundary on 22 February 2025 and the start of testing on 6 March 2025. Each record is counted by its earliest recorded stage start. (b) Dates, candidate records, exclusions and retained sample sizes of each partition.

Model development uses four expanding training blocks with validation weeks beginning on 25 January, 1 February, 8 February and 15 February 2025. Training sizes are 705, 924, 1,121 and 1,334 records and validation sizes 208, 187, 205 and 200, giving 800 common development targets over 28 observed origin dates. Training outcomes must be completed before each validation period starts, and

validation outcomes before the corresponding week ends, which keeps the expanding blocks separated. Assessment thus follows the temporal direction of model use and keeps future outcomes out of both model fitting and historical contexts [25, 26, 27, 35].

## 2.6. Reference baselines and current-input models

Three simple operational references were included. With  $\ell_i$  the pelleting line of operation  $i$ , the line-specific SEC reference estimates one energy intensity per line from training totals:

$$\hat{s}_\ell = \frac{\sum_{i \in \mathcal{T}: \ell_i = \ell} y_i}{\sum_{i \in \mathcal{T}: \ell_i = \ell} m_i}, \quad \hat{y}_i^{\text{SEC}} = m_i \hat{s}_{\ell_i}. \quad (3)$$

A line-specific median-energy reference and a recent-SEC reference, the median of the last five completed same-line energy-to-mass ratios, complete the set. The recent reference updates as outcomes become available during validation; the static references and the fitted models keep their parameters.

Let  $z_i$  be the transformed current-record inputs. Ridge regression estimates an unpenalized intercept and regularized coefficients,

$$(\hat{b}, \hat{\beta}) = \arg \min_{b, \beta} \left\{ \sum_{i \in \mathcal{T}} (y_i - b - z_i^\top \beta)^2 + 10 \|\beta\|_2^2 \right\}, \quad \hat{y}_i = \max(0, \hat{b} + z_i^\top \hat{\beta}). \quad (4)$$

The non-negative truncation prevents negative energy estimates without imposing a complete process-energy balance.

ExtraTrees, histogram gradient boosting and a multilayer perceptron (MLP) use the same current-record inputs with fixed configurations: 250 trees with a minimum leaf size of five for ExtraTrees; 200 boosting iterations, seven leaves, a learning rate of 0.05, a minimum leaf size of 20 and regularization of 10 for gradient boosting; and hidden layers of 32 and 16 units, Adam optimization, regularization of 0.1 and at most 800 iterations for the MLP. Neural predictions average seeds 11, 29 and 47; the tree methods use seed 2026. The configurations were held constant across validation blocks and represent the implementations evaluated here, not an exhaustive optimization of each family. When Ridge and gradient boosting gave practically the same development MAE, Ridge was chosen for parsimony, stability and record-level traceability rather than for predictive superiority. The selected Ridge model and its interval calibration were frozen before the 798 test records were evaluated.

For interpretation, the 22 design columns are assigned to eight groups: dosed mass, dosing-batch count, origin hour, origin weekday, pelleting line, product subfamily, presentation and bag/bulk category. The untruncated estimate and the contribution of group  $g$  are

$$f_i^{\text{Ridge}} = \widehat{b} + \sum_{g \in \mathcal{G}} C_{ig}, \quad C_{ig} = \sum_{k \in g} z_{ik} \widehat{\beta}_k, \quad \widehat{y}_i = \max(0, f_i^{\text{Ridge}}). \quad (5)$$

The grouped contributions preserve the arithmetic of the fitted model. They describe how the encoded variables build an estimate conditional on that model; they are not separate physical energy uses or causal effects [36, 37, 38].

### 2.7. Short-context temporal models on irregular operational events

The exploratory temporal comparison asks whether histories of completed same-line operations add information beyond the current-record covariates. Each eligible target receives the 16 most recent completed records from the same pelleting line and validation interval,

$$\mathcal{H}_i^{(16)} = \text{last}_{16, (a_j, j)} \{j : \ell_j = \ell_i, t_j \in \mathcal{P}, a_j < t_i, a_j < \sup \mathcal{P}\}, \quad (6)$$

where  $a_j$  is the recorded completion time and ties are broken by source-row identifier. An operation that started earlier but was not complete at the target origin cannot enter the context.

The 16-record window gives every model the same short-context representation and retains 584 of the 800 development targets: 149 from Pellet 1, 257 from Pellet 2 and 178 from Pellet 3 over 21 origin dates. Contexts restart at the weekly validation boundaries so that the expanding blocks remain separated; completed outcomes may inform later targets within the same week but never update fitted parameters. Figure 3 shows the expanding design and one observed context.

Two Ridge controls separate historical information from model capacity. The first uses only the current-record covariates; the second flattens the current covariates, the 16 historical covariate vectors and their energy outcomes, keeping the penalty of 10. TFT and N-HiTS receive the same current and historical information through their NeuralForecast implementations [39, 40]: TFT uses a hidden dimension of 32, four attention heads and dropout of 0.1, and N-HiTS uses three blocks with two 64-unit dense layers each, pooling factors of 2, 2 and 1 and interpolation downsampling factors of one. Each neural family uses seeds 11, 29 and 47, with an inner validation week immediately before each outer validation week. Optimization minimizes standardized-target MSE with Adam at a learning rate of  $10^{-3}$ , weight

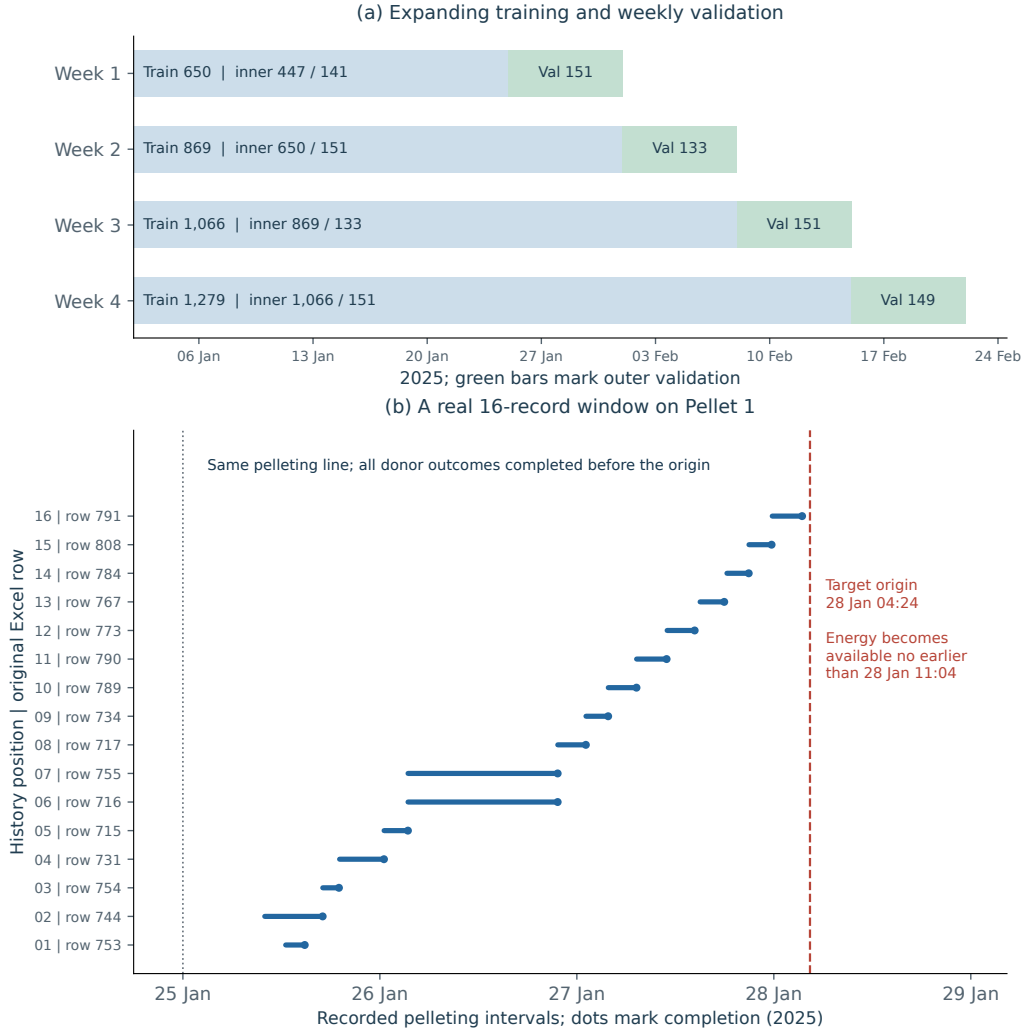


Figure 3: Short-context temporal evaluation. (a) Expanding training blocks and weekly validation blocks; green bars mark the outer validation weeks, and each training bar reports the earlier training / preceding-week validation counts used for epoch selection. (b) An observed context of 16 completed same-line operations for target row 823 on Pellet 1, ordered by completion; segments span recorded pelleting start and finish, dots mark completion, and every donor outcome precedes the target origin on 28 January at 04:24.

decay of  $10^{-4}$ , batch size 64 and gradient clipping at one, for at most 40 epochs with patience of eight; each seed is then refitted on the complete outer-training block for its internally selected number of epochs, and the three clipped predictions are averaged.

TimesFM 2.5 uses its frozen 200M checkpoint with energy-only input and with its native XReg adapter [22, 41, 42]; XReg fits a local covariate regression for each origin from its 16 historical records and adds the frozen TimesFM residual forecast. Chronos-2 is evaluated with energy-only and native-covariate inputs, using its median forecast with cross-query learning disabled [43, 44]. No foundation-model weights are updated locally. All temporal and control variants are evaluated on the same 584 targets, so the comparison concerns these specific short-context implementations on irregular pelleting events and is not a general ranking of the model families [45, 46].

## 2.8. Post-hoc exploratory extensions

Three post-hoc extensions turn patterns seen during development into hypotheses for later periods. H1 lets the mass association vary by pelleting line, H2 lets it vary by product subfamily, and H3 corrects the frozen Ridge estimate with recent same-line residuals. For H1 and H2, with grouping variable  $g_i$ , the untruncated predictor is

$$f_i = b + z_i^\top \beta + z_{m,i} \sum_{c \in C} \gamma_c \mathbf{1}(g_i = c), \quad (7)$$

with the same Ridge penalty and transformations fitted on training data only. H3 uses up to 20 recent same-line residuals of the frozen model, each eligible after its operation's recorded completion and ordered by completion time and then by source-row identifier, which gives a deterministic rule for tied completions. With  $e_j = y_j - \hat{y}_j^{\text{Ridge}}$  and  $n_i = |\mathcal{H}_i|$ ,

$$\delta_i = \begin{cases} \frac{n_i}{n_i + 10} \frac{1}{n_i} \sum_{j \in \mathcal{H}_i} e_j, & n_i \geq 5, \\ 0, & n_i < 5, \end{cases} \quad \hat{y}_i^{\text{H3}} = \max(0, \hat{y}_i^{\text{Ridge}} + \delta_i). \quad (8)$$

H1–H3 are exploratory development analyses: they generate candidate refinements for future chronological evaluation and do not replace the frozen model.

## 2.9. Empirically calibrated residual intervals

Absolute errors from the 380 calibration records calibrate residual intervals for the frozen model. For miscoverage level  $\alpha$ ,

$$q_\alpha = r_{(\lceil (n_{\text{cal}}+1)(1-\alpha) \rceil)}, \quad r_j = |y_j - \hat{y}_j|, \quad I_i^{1-\alpha} = [\max(0, \hat{y}_i - q_\alpha), \hat{y}_i + q_\alpha]. \quad (9)$$

The 90% and 95% half-widths are 93.01 and 132.96 kWh. The intervals are judged empirically by observed coverage, mean width and interval score, globally and by line; they are called residual intervals because the chronological industrial sequence is not assumed to be exchangeable. A post-hoc rolling extension updates the quantile from the latest 200 available absolute errors, starting from the calibration residuals and then admitting completed test outcomes while the point-prediction model stays frozen. Every donor satisfies

$$a_j < t_i, \quad a_j = \text{recorded pelleting completion of donor } j. \quad (10)$$

The rolling calculation is an exploratory adaptation comparator, not an independently validated improvement [47, 48, 49].

## 2.10. Performance metrics and statistical analysis

Point performance is measured by

$$\text{MAE} = N^{-1} \sum_i |\hat{y}_i - y_i|, \quad \text{RMSE} = \sqrt{N^{-1} \sum_i (\hat{y}_i - y_i)^2}, \quad (11)$$

$$\text{Bias} = N^{-1} \sum_i (\hat{y}_i - y_i), \quad \text{WAPE} = \frac{\sum_i |\hat{y}_i - y_i|}{\sum_i y_i}. \quad (12)$$

MAE is the primary selection criterion; RMSE emphasizes larger errors, bias quantifies systematic over- or underestimation, and WAPE expresses total absolute error relative to recorded energy [50]. Results are also reported by pelleting line and chronological period. Residual intervals are assessed by observed coverage, mean width and the interval score [31],

$$\text{IS}_\alpha(L, U; y) = (U - L) + \frac{2}{\alpha}(L - y)\mathbf{1}(y < L) + \frac{2}{\alpha}(y - U)\mathbf{1}(y > U). \quad (13)$$

After an operation completes, its signed excess and interval flag are

$$r_i^{\text{excess}} = y_i - \hat{y}_i, \quad B_i = \mathbf{1}\{y_i < L_i \text{ or } y_i > U_i\}. \quad (14)$$

These quantities support record-level review; they are statistical indicators and do not by themselves identify the physical cause of a deviation.

Pairwise differences use a circular block bootstrap with blocks of three observed origin dates, keeping all lines recorded on the same date in the same block, which preserves within-date dependence and short-range temporal grouping: 3,000 replicates for the primary test comparison and 5,000 for the exploratory comparisons. Alongside marginal 95% intervals, H1–H3 use approximate Bonferroni-adjusted 98.33% intervals and the seven temporal contrasts 99.29% intervals; the adjustments apply to the stated comparisons within each analysis family. Line- and week-specific results are reported with their sample sizes as stratified assessments.

### *2.11. Computational implementation*

Ridge and the native JAX XReg solver ran on CPU. Neural computations used float32 precision on an NVIDIA RTX PRO 5000 Blackwell GPU with 48 GB of memory, with TF32 disabled. TFT and N-HiTS were trained locally, whereas TimesFM and Chronos weights remained frozen. CPU–GPU numerical comparisons are computational diagnostics and are reported in Appendix [Appendix A](#).

## **3. Results**

### *3.1. Production variability across lines and periods*

Figure 4 summarizes production and electricity on the three lines: panels (a)–(c) contain the 2,742 records with positive pelleting energy, and panels (d)–(f) describe the mass, dosing-batch and product-family distributions of all 2,745 source records. Electricity generally rises with dosed mass, yet records of similar mass remain widely dispersed, and the distributions of mass, batch count and product family differ among lines and chronological partitions. Both observations argue for a production-adjusted baseline rather than a single plant-wide intensity value. Line-level differences in absolute or specific consumption must be read together with the corresponding production allocation; they are not controlled comparisons and say nothing about intrinsic mechanical efficiency.

### *3.2. Baseline selection and frozen chronological test*

Table 3 gives the development results of the seven candidates on the same 800 records. Ridge and histogram gradient boosting were practically tied at 51.48 and 51.50 kWh MAE, with ExtraTrees close behind at 52.31 kWh; the MLP and the three references had larger errors, and the recent-SEC estimator, although updated with completed validation outcomes, did worse than the static line-specific SEC.

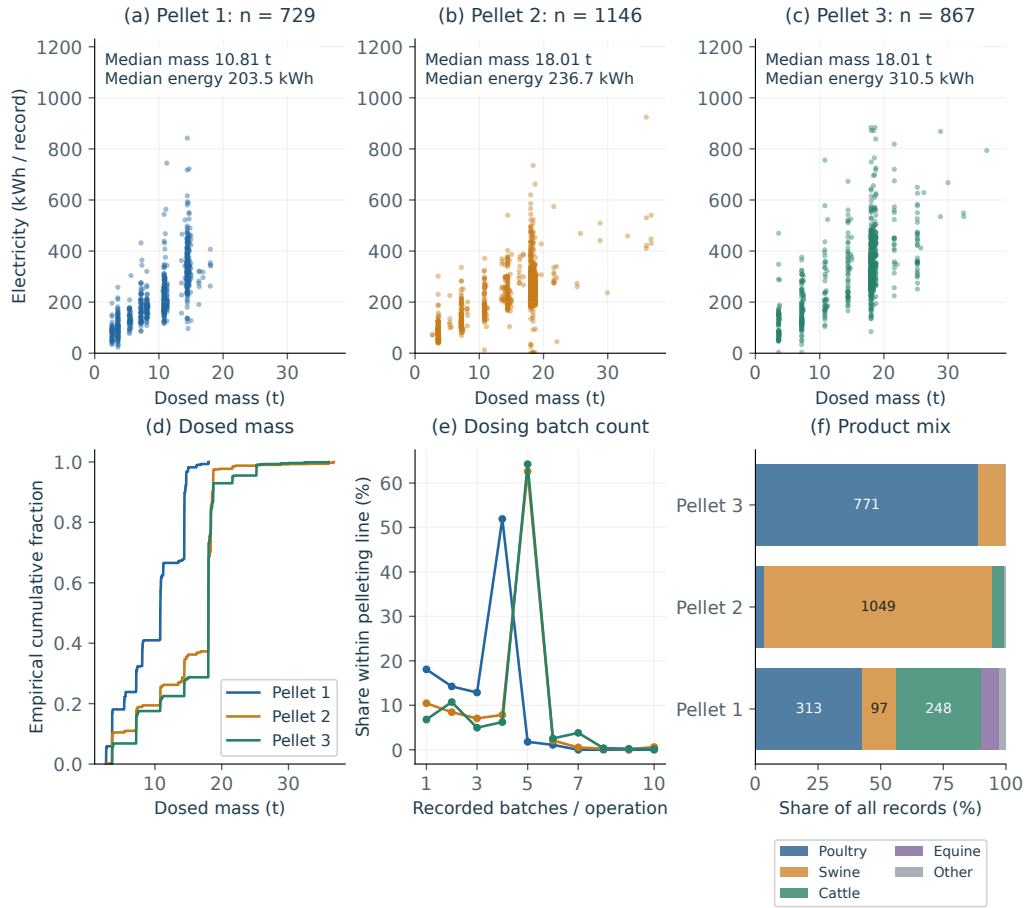


Figure 4: Production and electricity across pelleting lines. (a–c) Dosed mass and pelleting electricity for the 2,742 positive-energy records, with line-specific medians annotated. (d–f) Mass distributions, dosing-batch frequencies and product-family mix of all 2,745 source records; values inside the product bars are record counts. Points are individual operations and are not uniformly spaced in time.

Ridge was selected for comparable accuracy, lower structural complexity and the availability of an exact additive decomposition, not because it was shown to be more accurate than gradient boosting.

Table 3: Chronological development performance on the same 800 records. The recent-SEC estimator was updated as completed outcomes became available during validation; all other models kept fixed parameters within each validation block.

| Method                        | MAE (kWh) | RMSE (kWh) | Bias (kWh) | WAPE (%) |
|-------------------------------|-----------|------------|------------|----------|
| Ridge                         | 51.48     | 72.31      | −3.14      | 19.88    |
| Gradient boosting             | 51.50     | 73.46      | −5.72      | 19.88    |
| ExtraTrees                    | 52.31     | 75.37      | −4.89      | 20.20    |
| MLP, three-seed mean          | 56.64     | 81.36      | −1.47      | 21.87    |
| Line-specific SEC             | 59.69     | 83.92      | −6.59      | 23.05    |
| Recent SEC, last five records | 73.51     | 109.55     | 4.29       | 28.38    |
| Line-specific median          | 82.61     | 112.78     | −14.09     | 31.89    |

The selected specification was fitted on the 1,542 eligible development records and frozen before the 798-record chronological test. Ridge reduced test MAE from 61.29 to 52.45 kWh (Table 4), a paired reduction of 8.84 kWh per record or 14.42% with a 95% block-bootstrap interval of [6.08, 11.50] kWh; RMSE fell from 92.05 to 80.42 kWh and global WAPE from 22.34% to 19.11%. These percentages are reductions in prediction error, not in plant electricity consumption. Removing the five test records longer than 24 hours gave 52.32 kWh for Ridge and 61.15 kWh for SEC, so those records do not explain the advantage.

Table 4: Frozen chronological test performance overall and by pelleting line. Global WAPE is 19.11% for Ridge and 22.34% for line-specific SEC.

| Population | Method            | <i>N</i> | MAE (kWh) | RMSE (kWh) | Bias (kWh) |
|------------|-------------------|----------|-----------|------------|------------|
| All lines  | Ridge             | 798      | 52.45     | 80.42      | −21.89     |
| All lines  | Line-specific SEC | 798      | 61.29     | 92.05      | −24.57     |
| Pellet 1   | Ridge             | 200      | 66.37     | 105.14     | −53.85     |
| Pellet 1   | Line-specific SEC | 200      | 72.31     | 113.90     | −64.03     |
| Pellet 2   | Ridge             | 333      | 43.36     | 61.26      | −10.85     |
| Pellet 2   | Line-specific SEC | 333      | 49.29     | 73.24      | −8.66      |
| Pellet 3   | Ridge             | 265      | 53.36     | 80.10      | −11.63     |
| Pellet 3   | Line-specific SEC | 265      | 68.05     | 94.79      | −14.79     |

The production-adjusted model had lower MAE than SEC on every line (Figure 5). Performance was most consistent on Pellet 2 and Pellet 3, with Ridge MAE of 43.36 and 53.36 kWh. Pellet 1 also benefited, its MAE falling from 72.31 to 66.37 kWh and its bias from  $-64.03$  to  $-53.85$  kWh, but the remaining systematic underestimation means that the frozen model was not operationally calibrated for that line during the test period. This localized shortfall does not invalidate the baseline or the gains on the other lines; it does mean that a common alerting criterion should not be applied across lines without line-specific calibration and chronological validation.

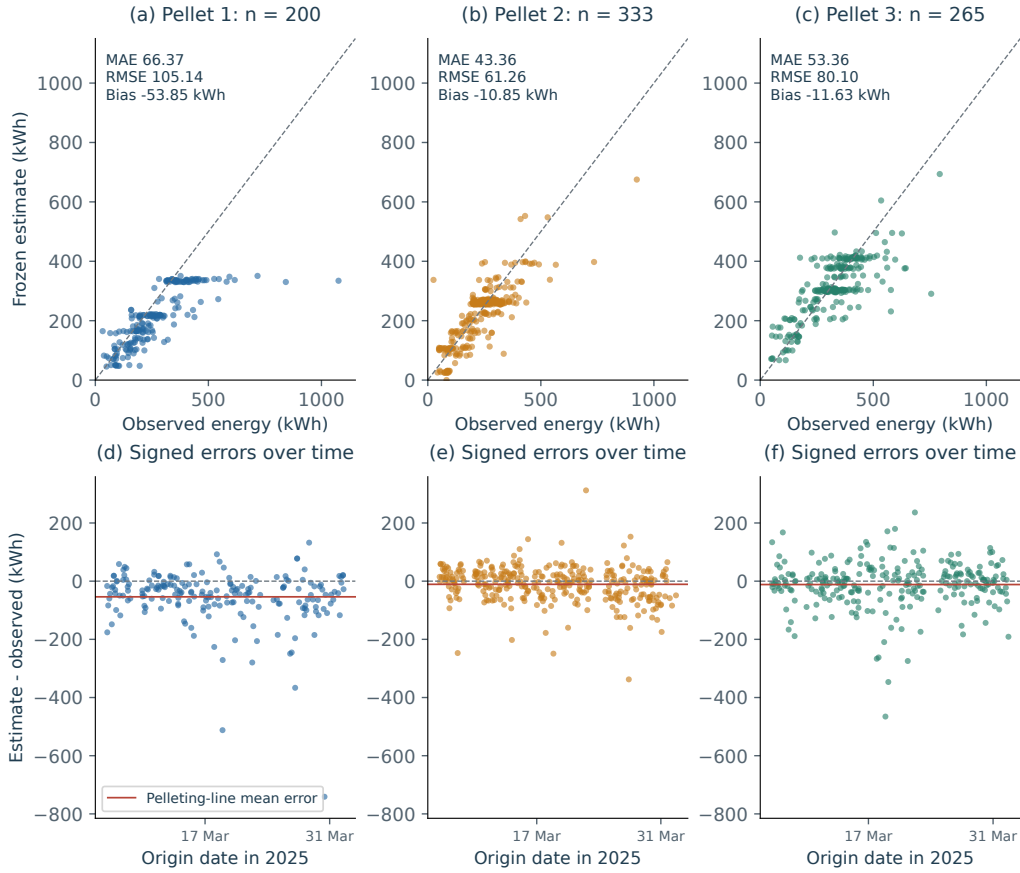


Figure 5: Frozen Ridge performance in the chronological test. (a–c) Observed versus predicted electricity on identical axes, with the equality line dashed and MAE, RMSE and bias in kWh. (d–f) Signed errors in chronological order within each line, with zero dashed and the mean signed error of each line in red. The 200, 333 and 265 records of Pellet 1, 2 and 3 make up all 798 test observations.

### 3.3. Predictive value of completed operating histories

The short-context comparison uses the 584 development records for which 16 completed same-line operations were available within the weekly interval; the other 216 of the original 800 did not meet the requirement, so these results apply to the matched population and are not comparable with the 800-record ranking. Table 5 separates the Ridge controls, the locally trained temporal models and the frozen pretrained models.

Table 5: Short-context model performance on the 584 matched development records. TFT and N-HiTS are three-seed prediction means; foundation-model weights remain frozen.

| Model group       | Method                   | MAE    | RMSE   | Bias   | WAPE (%) |
|-------------------|--------------------------|--------|--------|--------|----------|
| Ridge controls    | Current inputs           | 53.55  | 77.01  | −4.72  | 20.37    |
|                   | Full 16-record window    | 63.95  | 85.95  | −1.42  | 24.32    |
| Local temporal    | TFT                      | 53.65  | 79.11  | −11.23 | 20.41    |
|                   | N-HiTS                   | 65.28  | 92.56  | −3.05  | 24.83    |
| Frozen pretrained | Chronos-2 + covariates   | 85.53  | 120.82 | −8.11  | 32.53    |
|                   | TimesFM 2.5, energy only | 96.40  | 130.72 | −9.22  | 36.67    |
|                   | Chronos-2, energy only   | 100.02 | 136.47 | −11.60 | 38.04    |
|                   | TimesFM 2.5 + XReg       | 112.12 | 644.61 | 28.03  | 42.65    |

MAE, RMSE and bias are in kWh per record.

Current-input Ridge reached 53.55 kWh MAE and TFT 53.65 kWh. The paired difference (Ridge minus TFT) was −0.11 kWh with a 95% interval of [−2.20, 2.32] kWh, which includes effects in both directions and gives no support for an improvement from TFT in the tested configuration. Giving Ridge the full 16-record history raised its MAE to 63.95 kWh, and N-HiTS reached 65.28 kWh. Chronos-2 did better with covariates than with energy alone, but both versions stayed well above the Ridge control, as did the energy-only TimesFM variant (Figure 6).

The TimesFM XReg configuration produced three extreme Pellet 3 predictions of about 2,753, 9,239 and 12,203 kWh for observed values of 185.4, 142.4 and 70.9 kWh. In those contexts dosed mass had almost no historical variability, so the standardized current mass extrapolated far outside the local range; an independent float64 reconstruction reproduced the behaviour, and the three predictions are retained in the official metrics. The diagnosis concerns numerical instability of this short-context XReg pipeline and says nothing about TimesFM with longer contexts or other scaling and regularization policies. Results by validation week,

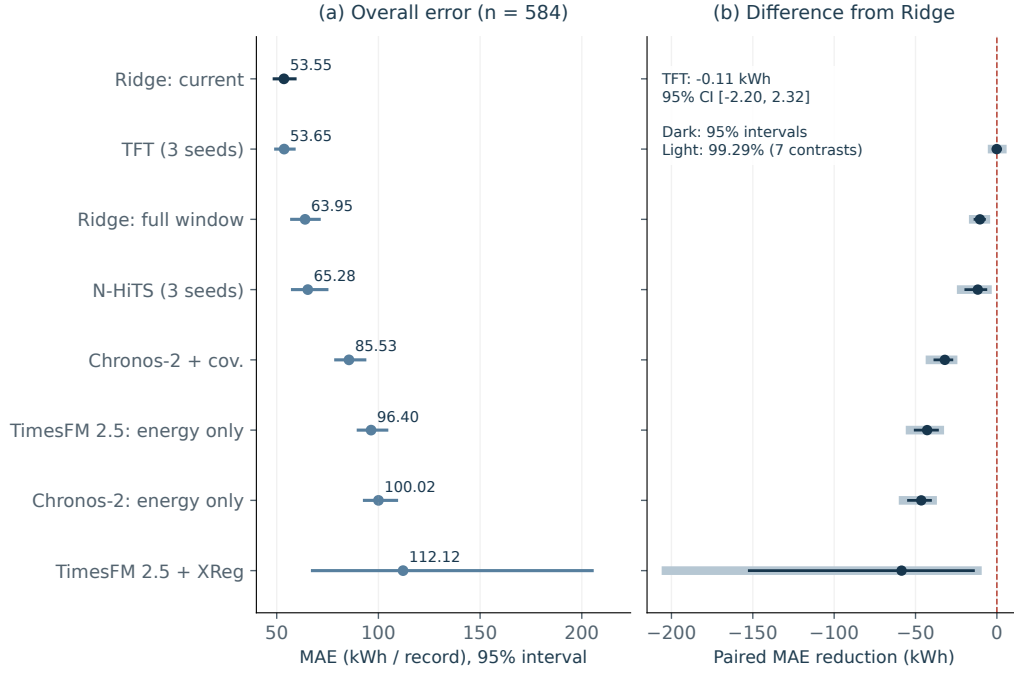


Figure 6: Short-context comparison on the 584 matched development records. (a) MAE with 95% block-bootstrap intervals. (b) Paired MAE reduction relative to current-input Ridge, so positive values favour the alternative; pale intervals are adjusted for seven comparisons and dark intervals are unadjusted 95% intervals. The wide TimesFM XReg interval is shown in full.

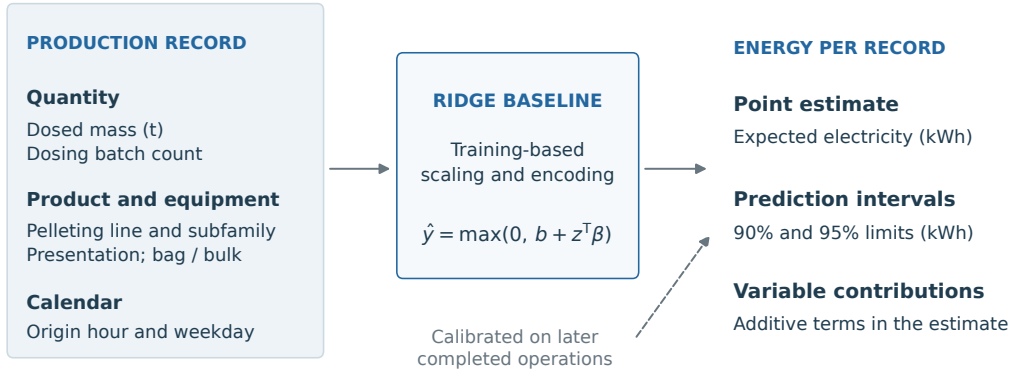
by line and by seed, together with the paired intervals of every contrast, are given in Supplementary Tables S12–S16 and Figs. S3–S4. Under the tested 16-record, same-line, within-week representation, historical inputs did not improve prediction over current-record covariates. This does not imply that operating histories lack value under other context definitions, longer observation periods or prospectively selected temporal models.

### 3.4. Additive decomposition of the fitted predictor

Figure 7 shows the operation-level output of the baseline: panel (a) the transformation of recorded production characteristics into an expected electricity value, a residual interval and a signed deviation, and panel (b) these outputs for source row 2080, a Pellet 2 operation with 18.002 t distributed over five dosing batches. The model estimated 255.90 kWh; the recorded value was 267.30 kWh, a residual of 11.40 kWh inside the fixed 95% interval of [122.94, 388.85] kWh. The

record was chosen because its observed energy is close to the median test value; it does not represent the whole error distribution.

#### a From production conditions to an energy baseline



#### b Reading the output: an actual operation

##### Pellet 2 | 10 March 2025

18.002 t; 5 dosing batches  
Swine-finishing product  
Pelleted; bagged  
Origin 08:31; completion 11:21

Source row 2080

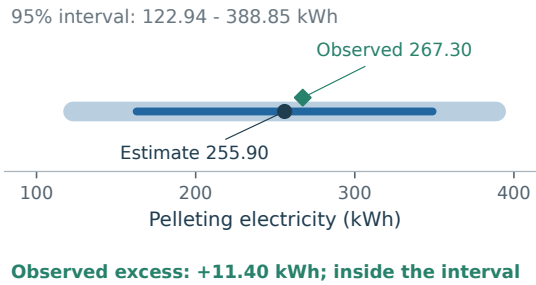


Figure 7: Production-adjusted baseline output. (a) Recorded production conditions are transformed into an expected electricity value, residual intervals and a signed deviation; residuals from the later calibration period set the interval limits. (b) A real Pellet 2 operation (source row 2080): the light segment is the fixed 95% interval, the dark segment the 90% interval, the circle the estimate and the diamond the observed energy. An excess above the estimate is a model residual, not a measured saving or a confirmed fault.

Applying Equation 5 to every frozen estimate reproduced all 798 archived test values to within  $1.2 \times 10^{-13}$  kWh, and none required non-negative clipping. Figure 8 separates the fitted coefficients from record-specific contributions: panel (a) gives the numerical coefficients in original input units, (b) the categorical coefficients under the fitted coding, (c) the distributions of the grouped contributions and (d) the decomposition of source row 2080, whose intercept and eight grouped contributions

sum exactly to 255.90 kWh. After undoing the standardization used in fitting, the mass coefficient is 13.30 kWh per additional tonne and the dosing-batch coefficient 9.52 kWh per additional batch, conditional on the other encoded predictors (all 22 coefficients and the conversion factors are listed in Supplementary Tables S6–S7).

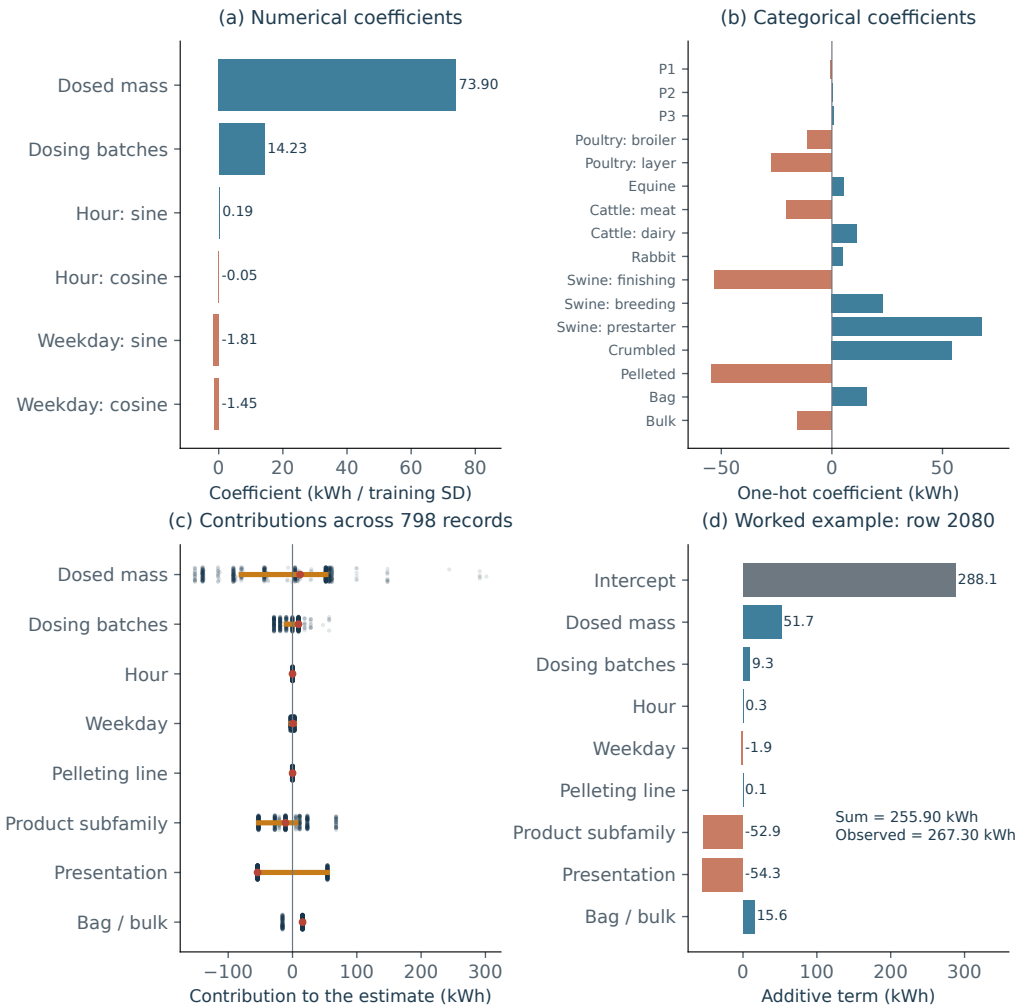


Figure 8: Additive decomposition of the frozen Ridge predictor. (a) Numerical coefficients in original input units. (b) All 16 categorical coefficients under the fitted coding. (c) Grouped contributions for every test record, with medians in red and interquartile ranges in orange. (d) Arithmetic decomposition of source row 2080. The reference combines standardized numerical means with all-zero categorical indicators and need not describe a feasible operating condition; contributions are conditional model terms, not causal effects.

These values are coefficients of a fitted linear predictor, not separately identified physical requirements. Mass and batch count are correlated, and line and product allocation were not randomized, so their signs and magnitudes describe conditional associations. The intercept of 288.09 kWh is the algebraic reference of the fitted coding, not a standby consumption; categorical coefficients depend on the one-hot representation and its reference configuration; and small line coefficients do not demonstrate equal intrinsic efficiency among the lines. What the exact equality does provide is record-level traceability, without assigning causal meaning or physically separable energy uses to individual terms.

### 3.5. *Post-hoc exploratory extensions*

Three post-hoc extensions were evaluated on the 800 development records (Table 6). Letting the mass association vary by product subfamily reduced MAE by 1.44 kWh (2.80%) and the recent-residual correction by 1.71 kWh (3.33%), both with multiplicity-adjusted intervals above zero; the line-dependent mass interaction reduced MAE by only 0.22 kWh, with an interval crossing zero. The gains varied across lines and validation weeks (Supplementary Fig. S1 and Tables S8–S9). Because H1–H3 were motivated by patterns observed during development, they are hypothesis-generating and need evaluation in a new chronological period; they do not replace the specification frozen for the primary test.

Table 6: Post-hoc exploratory point-model extensions on the 800 development records. Positive reductions favour the extension; the adjusted intervals cover these three comparisons only.

| Variant                    | MAE   | MAE reduction | Adjusted 98.33% interval |
|----------------------------|-------|---------------|--------------------------|
| H1: mass by pelleting line | 51.26 | 0.22          | [−0.08, 0.58]            |
| H2: mass by subfamily      | 50.04 | 1.44          | [0.55, 2.48]             |
| H3: recent residuals       | 49.77 | 1.71          | [0.24, 3.16]             |

All quantities are in kWh per record.

### 3.6. *Residual intervals and records flagged for review*

Figure 9 shows the frozen estimates and fixed 95% intervals for every test record, in chronological order within each line. The fixed 90% and 95% intervals covered 85.96% and 93.23% of the 798 records with mean widths of 183.63 and 258.85 kWh; their block-bootstrap 95% coverage intervals, [81.86, 90.24]% and [90.53, 95.87]%, make global coverage broadly compatible with the nominal levels. Line-specific coverage differed: at the nominal 95% level it was 90.00% on Pellet 1

(20 records outside the interval), 96.40% on Pellet 2 (12) and 91.70% on Pellet 3 (22).

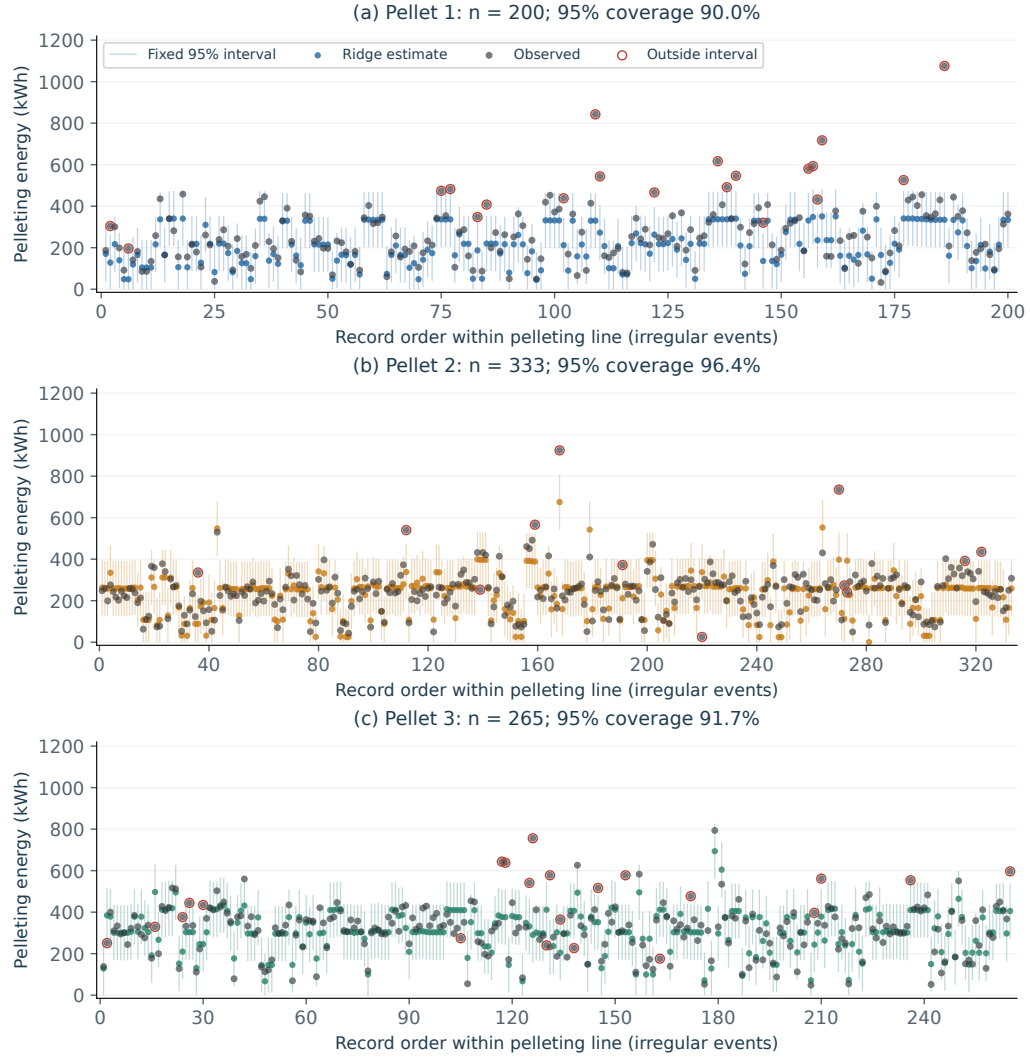


Figure 9: Frozen Ridge estimates and fixed 95% residual intervals for every test record, in chronological order within each pelleting line. Black points are observed electricity, coloured points the frozen estimate and vertical lines the intervals; red circles mark the 54 outcomes outside their intervals. Horizontal position is record order, so equal spacing does not imply equal elapsed time.

In total 54 records fell outside the fixed 95% intervals, 45 above the upper limit and nine below the lower one: about 6.77 flagged records per 100 operations, 5.64

upper and 1.13 lower exceedances. They are candidates for technical review because the calibrated baseline did not represent their energy well; without independent plant evidence they should not be classed as confirmed anomalies, inefficient operations, equipment faults or avoidable consumption.

The post-hoc rolling policy, which updates the quantiles from the most recent 200 completed residuals, raised coverage to 89.47% and 94.49% but widened the intervals by 19.66% and 18.85% while improving the interval score by only 1.12% and 0.23% (Table 7); most of the additional coverage came from width. The number of records outside the 95% interval fell from 54 to 44. Because the policy was defined after the test results had been inspected and was not evaluated on an independent period, it is a calibration diagnostic, not a validated replacement for the fixed intervals. Coverage, width and interval score by line and by week for both policies are given in Supplementary Tables S10–S11 and Fig. S2.

Table 7: Fixed and post-hoc rolling residual intervals on the same 798 frozen point predictions. The rolling policy has not been validated on an independent period.

| Policy                 | Nominal level | Coverage (%) | Mean width | Interval score |
|------------------------|---------------|--------------|------------|----------------|
| Fixed                  | 90%           | 85.96        | 183.63     | 381.79         |
| Rolling, 200 residuals | 90%           | 89.47        | 219.74     | 377.51         |
| Fixed                  | 95%           | 93.23        | 258.85     | 497.64         |
| Rolling, 200 residuals | 95%           | 94.49        | 307.64     | 496.48         |

Width and interval score are in kWh per record.

## 4. Discussion

### 4.1. Production adjustment against intensity-based references

During development, Ridge produced lower prediction errors than the line-specific median, recent SEC, MLP, ExtraTrees and line-specific SEC references and performed like histogram gradient boosting; in the frozen chronological test its MAE was 14.42% below line-specific SEC. This agrees with energy-baseline research that recommends adjusting consumption for the relevant operating conditions rather than relying on aggregate or intensity-based indicators [13, 14, 15, 16], and it extends the pelleting studies that link energy demand to production volume, formulation, conditioning, equipment and operating conditions [11, 8, 9, 10, 12]. The contribution is a chronologically evaluated baseline that combines production

mass, product characteristics and line identity for heterogeneous completed operations, which is the kind of normalization that ISO 50006 and the IPMVP expect of an energy baseline [5, 6]. The improvement is in prediction accuracy: it is neither a reduction in actual energy use nor an estimate of achievable savings, and Ridge was chosen for its balance of accuracy, parsimony and traceability rather than for universal superiority over nonlinear models.

#### 4.2. *Line-specific behaviour*

Production adjustment helped on all three lines. Pellet 2 and Pellet 3 showed the most consistent agreement between predictions and observations and are the strongest basis for prospective validation. Pellet 1 also improved on line-specific SEC, but its remaining negative bias shows that the frozen model was not operationally calibrated for that line. This localized requirement does not undermine the common modelling framework; it shows that aggregate performance can hide line-specific deficiencies. Differences among lines may reflect equipment configuration, production allocation, maintenance, metering or unrecorded operating conditions and should not be read as intrinsic differences in energy efficiency.

#### 4.3. *Operating histories, temporal models and foundation models*

Under the tested 16-record, same-line, within-week representation, completed-operation histories did not improve on current-record covariates. Dosed mass, product characteristics and line identity may already capture much of the repeatable structure, and irregular intervals weaken the meaning of sequential proximity. The conclusion is confined to the evaluated short-context representation and does not contradict the value of temporal models where sequences carry informative dependence [20, 21]; likewise, the TimesFM and Chronos results characterize the tested interfaces, not the general capabilities of foundation models [22, 23, 43]. The XReg extrapolation and the device-dependent normalization sensitivity of Chronos-2 (Appendix [Appendix A](#)) show that covariate adaptation and normalization shape the final industrial prediction as much as the pretrained weights do, and that a model’s ability to accept the available covariates is no evidence that its pipeline is stable for this kind of record. Longer contexts, explicit elapsed time, operating-state information, maintenance events and regularly sampled measurements would allow a fairer assessment of temporal information in industrial pelleting [45, 46].

#### 4.4. *Reading the additive decomposition*

Ridge gives an exact additive decomposition of its fitted linear predictor and therefore a direct, reproducible audit trail. Its coefficients and record-specific contributions nevertheless depend on scaling, categorical coding, reference categories,

feature construction and correlations among predictors [36, 37, 38]: they describe the fitted prediction function, do not identify causal effects and would change under alternative coding choices. Line and product terms may also absorb unrecorded differences in formulation, equipment condition, production allocation or operating practice. The small exploratory gains from product-dependent mass effects and residual correction indicate where the next refinement may come from, but they are hypotheses for a new period, not reasons to replace the frozen model on the strength of an already examined test.

#### *4.5. Uncertainty and the meaning of a flagged record*

The residual intervals showed that satisfactory marginal coverage does not guarantee comparable line-specific calibration [33, 32, 34]. Pellet 2 had the most stable coverage, whereas Pellet 1 and, to a lesser extent, Pellet 3 need further assessment. Rolling calibration raised coverage mainly by widening the intervals, which confirms that coverage must be read together with interval width, bias and alert frequency [30, 31]. An interval exceedance is therefore a candidate for technical review, not a confirmed anomaly, inefficiency, equipment failure or energy-saving opportunity.

#### *4.6. Scope of use and digital-twin integration*

The evidence supports retrospective assessment after each recorded operation closes: an expected electricity value, an empirically calibrated residual interval and a traceable deviation that identifies candidates for review. Prospective supervisory alerts require validation over a new consecutive production period with prespecified line-level criteria and, for Pellet 1, additional recalibration or predictors. Automatic control and modification of process set-points remain outside the evidence, because the observational model does not estimate causal responses. A digital-twin application should therefore begin with validated monitoring and decision support [51], storing the input and model version of each issued estimate and attaching the measured outcome when it becomes available, so that an earlier forecast and a later production-adjusted assessment remain distinguishable.

#### *4.7. Strengths and limitations*

The study draws its strength from real industrial records, chronological separation, operational comparators, line-specific diagnostics, residual intervals and a reproducible decomposition. Its transferability is limited by the absence of a verified production-order identifier, retained temporal overlaps, unverified meter boundaries and undocumented predictor-availability times. The data cover three lines of

one plant over about three months, use product categories rather than complete formulations, and lack detailed maintenance and operating-state variables. The intervals mainly assess marginal coverage, several extensions were post hoc, and the operational records cannot be released publicly, although they are available from the corresponding author on reasonable request. The contribution therefore lies in the evaluation framework—production adjustment, chronological testing, line-specific assessment, uncertainty quantification and algebraic traceability—rather than in a universally deployable fitted model.

## 5. Conclusions

A production-adjusted Ridge model gave a more accurate and more traceable baseline for pelleting electricity than the operational references evaluated here. In the frozen chronological test it reduced MAE by 14.42% relative to line-specific SEC while providing a reproducible additive decomposition of every estimate. The improvement concerns the estimation of expected electricity under the recorded production conditions; it is not a reduction in plant energy use or evidence of achievable savings.

Under the tested 16-record, same-line, within-week representation, completed operating histories did not improve on current-record covariates, and neither TFT, N-HiTS nor the tested TimesFM 2.5 and Chronos-2 configurations beat the current-input control. Production adjustment helped on all three lines, most consistently on Pellet 2 and Pellet 3; Pellet 1 improved on SEC, but its systematic underestimation calls for line-specific recalibration or additional predictors before prospective use.

The evidence supports retrospective assessment at the closure of recorded operations, where the baseline provides an expected value, an empirically calibrated residual interval and a traceable deviation for identifying candidates for technical review. Prospective supervisory use requires validation on a new consecutive production period with prespecified line-level criteria and verified record identity, measurement boundaries, predictor availability and calibration rules. Until then, automatic control, modification of process set-points and operational digital-twin integration remain outside the demonstrated scope of the study.

## CRedit authorship contribution statement

**Vladimir Sousa Santos:** Conceptualization, Methodology, Investigation, Writing – original draft, Project administration, Funding acquisition. **Juan José Cabello Eras:** Conceptualization, Supervision, Resources, Writing – review &

editing. **Eder Elías Herrera Rodiño:** Investigation, Data curation, Resources, Writing – review & editing. **Francisco Manuel Arrabal-Campos:** Methodology, Software, Formal analysis, Data curation, Validation, Visualization, Writing – original draft, Writing – review & editing.

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## **Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## **Supplementary material**

Supplementary material associated with this article (the full source-variable dictionary, descriptive statistics, the complete coefficient table of the frozen Ridge model, bootstrap and calibration details, residual-interval diagnostics by line and week, stratified and per-seed results of the temporal benchmark, CPU–GPU computational diagnostics and four additional figures) is provided as a separate document.

## Data availability

The code, frozen model configurations, archived aggregate results, figure scripts and LaTeX sources are openly available in the GitHub repository <https://github.com/fmarrabal/pelleting-energy-baseline-real-data> (release v4-submission). The industrial production and electricity records, and the record-level tables derived from them, belong to the plant and are not publicly available; they can be obtained from the corresponding author ([fmarrabal@ual.es](mailto:fmarrabal@ual.es)) on reasonable request, subject to the data owner’s agreement. Pretrained checkpoints are identified by repository and revision and are not redistributed.

## Appendix A. Computational and numerical reproducibility

The GPU experiment used Python 3.13.13, PyTorch 2.14.0 with CUDA 13.0, NeuralForecast 3.2.2, the TimesFM package 3.0.2, Chronos-forecasting 2.3.2 and scikit-learn 1.9.0. The pretrained checkpoint revisions were 1d952420fba87f3c6dee4f240de0f1a0fbc790e3 for TimesFM 2.5 and 29ec3766d36d6f73f0696f85560a422f50e8498c for Chronos-2. TimesFM received 16 observed context records, padded to a maximum context of 32, with a configured maximum horizon of 32; only its first forecast position was evaluated. Chronos used float32, disabled cross-query learning and returned the median forecast. Model configurations, checkpoints and record-level outputs are retained in the repository; pretrained weights are identified by revision rather than redistributed.

All 128,896 logged donor relationships passed the checks for same-pelleting-line membership, partition membership and completion before the target origin; the total includes repeated relationships across nested and expanding blocks. Source-data, frozen-model and final-prediction fingerprints remained unchanged, and the shared CPU and GPU data, window and encoding functions were checked for consistency. These checks concern the recorded completion-time proxy, since actual publication delays are not available in the source.

Within the GPU run, the 24 saved neural checkpoints were reloaded and compared with their stored inference, together with four Chronos covariate batch-size checks. All 28 checks passed the recorded tolerances with a maximum difference of  $7.63 \times 10^{-5}$  kWh. Fresh CPU and GPU neural fits were less closely aligned, as optimization can follow different trajectories even with the same integer seed: TFT’s CPU and GPU MAEs were 53.25 and 53.65 kWh, and neither established an advantage over the matched Ridge control under the paired intervals. Figure A.10 shows all three GPU seeds and their ensemble.

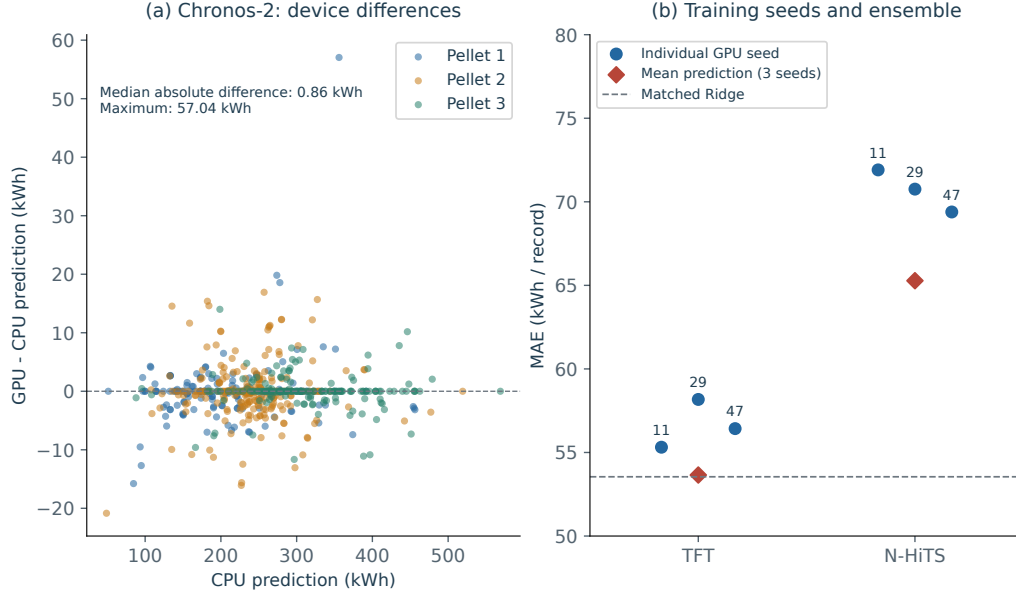


Figure A.10: Numerical and training variability. (a) Every paired CPU/GPU Chronos-2 covariate output as a signed difference against the CPU prediction. (b) All three GPU seeds (11, 29 and 47) and their prediction-mean ensemble for TFT and N-HITS, with the matched Ridge reference; averaging predictions is not the same as averaging the three MAEs. These diagnostics accompany the primary comparison without selecting a preferred seed or device.

For frozen Chronos-2 with covariates, the median absolute CPU–GPU prediction difference was 0.86 kWh and the maximum 57.04 kWh, and its overall MAE moved from 85.65 to 85.53 kWh. Maximum prediction differences for both frozen TimesFM variants and for univariate Chronos stayed below  $1.9 \times 10^{-4}$  kWh. To trace the covariate sensitivity, the installed Chronos preprocessing and instance normalization were applied to the same 584 contexts, giving 6,424 target/covariate channels. Among the 1,097 constant historical channels, 694 showed a CPU–GPU difference above 0.1 in normalized context values, whereas no non-constant channel exceeded that threshold; the largest difference was 0.88137 in the transformed coordinates.

The implementation substitutes a small scale only when the calculated scale is exactly zero, so tiny round-off variances in constant channels can be amplified before the inverse-hyperbolic-sine transform. This mechanism is consistent with the observed sensitivity, although it is not an end-to-end causal demonstration for each prediction. The reported benchmark keeps the native preprocessing and original

outputs; a corrective normalization policy would require a separately selected and evaluated experiment.

### **Declaration of generative AI and AI-assisted technologies in the writing process**

During the preparation of this work the authors used Trinka as a language-editing assistant to improve the readability and English of the manuscript, and Claude Code as a coding assistant for writing and documenting the analysis scripts. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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